Quantum Mechanics II Formulas & Methods

Idan Stark

February 1, 2026

1 Natrual Units

$$\hbar = c = 1$$

Mass Dimension - The power of the mass unit needed to represent the unit of the given quantity. For basic quantities:

$$[m] = [E] = [p] = 1 \qquad [x] = [t] = -1$$

Derivatives and integrals (vectors and covectors):

$$[\partial_\mu] = 1 \qquad [dx^\mu] = -1$$

Action, Lagrangian and Hamiltonian:

$$[S] = 0 \qquad [L] = [H] = 1 \qquad [\mathscr{L}] = [\mathscr{H}] = 4$$

2 Special Relativity

The Minkowski metric:

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

$$ds^2 = -dt^2 + d\vec{x}^2$$

$$x^\mu = \eta^{\mu\nu} x_\nu$$

2.1 Lorentz transformations

Boosts:

$$\Lambda_{\mu\nu} = \begin{pmatrix} \cosh a_x & \sinh a_x & & \\ \sinh a_x & \cosh a_x & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

Rotations:

$$\Lambda_{\mu\nu} = \begin{pmatrix} 1 & 0 & & \\ 0 & 1 & & \\ & & \cos\theta_x & \sin\theta_x \\ & & -\sin\theta_x & \cos\theta_x \end{pmatrix}$$

2.2 Electromagnetics

The EM potential:

$$A^\mu = (\phi, -\vec{A})$$

The EM Free Lagrangian:

$$\mathscr{L} = \frac{1}{4} F^{\mu\nu} F_{\mu\nu} \qquad F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

Which gives the classica EOM:

$$\Box A^\mu - \partial^\mu \partial_\nu A^\nu = 0$$

3 Quantizing a theory

- Starting from a Lagrangian density $\mathscr{L}$ (with allowed terms)
- Find Conjugate Momentum π to the field
- Derive EOM from E-L equations
- Solve the EOM to find the propating solutions
- Promote the fields and momenta to quantum operators by defining commutation relations
- Expand the fields in EOM solutions (mode expansion), promote their coefficient to quantum operators - the ladder operators.
- Check the H is non-negative and all states have a positive norm

4 Free theories

4.1 Real Scalar Field

Lagrangian and Hamiltonian:

$$\mathscr{L} = \frac{1}{2} \left(\partial_\mu \phi\right)^2 - \frac{1}{2} m^2 \phi^2 \qquad \mathscr{H} = \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2} m^2 \phi^2 \qquad \pi = \partial_0 \phi$$

Equation of Motion - The Klein Gordron equation

$$\left(\Box - m^2\right) \phi = 0 \qquad \Box = \partial_\mu \partial^\mu = -\partial_t^2 + \nabla^2$$

Solutions:

$$\phi(x) = \int \frac{d^4 p}{(2\pi)^4} e^{-ipx} \qquad \begin{array}{c} px = p^\mu x_\mu = p^0 t - \vec{p}\cdot\vec{x} \\ p^0 = \omega_{\vec{p}} = \sqrt{m^2 + \vec{p}^2} \end{array}$$

Commutation Relations (Bosonic):

$$\begin{gathered} [\phi_x, \pi_y] = \delta^3(\vec{x} - \vec{y}) \qquad [\phi_x, \phi_y] = [\pi_x, \pi_y] = 0 \\ \left[a_{\vec{k}}, a^\dagger_{\vec{p}}\right] = (2\pi)^3 \delta^3(\vec{k} - \vec{p}) \qquad [a_{\vec{k}}, a_{\vec{p}}] = \left[a^\dagger_{\vec{k}}, a^\dagger_{\vec{p}}\right] = 0 \\ [H, a_{\vec{k}}] = -\omega_{\vec{k}} a_{\vec{k}} \qquad \left[H, a^\dagger_{\vec{k}}\right] = \omega_{\vec{k}} a^\dagger_{\vec{k}} \end{gathered}$$

Mode Expansion

$$\phi(t,\vec{x}) = \int \frac{d^3p}{(2\pi)^3\sqrt{2\omega_p}}\left(a_{\vec{p}}e^{-ipx} + a^\dagger_{\vec{p}}e^{ipx}\right)$$
$$\pi(t,\vec{x}) = \partial_0\phi(t,\vec{x}) = i\int \frac{d^3p}{(2\pi)^3}\sqrt{\frac{\omega_p}{2}}\left(-a_{\vec{p}}e^{-ipx} + a^\dagger_{\vec{p}}e^{ipx}\right)$$
$$H = \int \frac{d^3p}{(2\pi)^3}\omega_p\left(a^\dagger_{\vec{p}}a_{\vec{p}} + \frac{1}{2}(2\pi)^3\delta^3(\vec{p}-\vec{p})\right)$$

4.2 Complex Scalar Field

Lagrangian:

$$\mathscr{L} = \partial_\mu\phi^*\partial^\mu\phi - m^2\phi^*\Phi$$

Two ways to analyse: consider ϕ and ϕ^* as two scalar fields or consider $\mathrm{Re}\{\phi\}$ and $\mathrm{Im}\{\phi\}$ as two scalar fields. Usully we follow the former.
Hamiltonian:

$$\mathscr{H} = \pi^*\pi + (\nabla\phi^*)(\nabla\phi) + m\phi^*\phi$$
$$\pi = \partial\phi^* \qquad \pi^* = \partial\phi$$

Equation of Motion - The Klein Gordron equation

$$\left(\Box - m^2\right)\phi = 0 \qquad \left(\Box - m^2\right)\phi^* = 0$$

Solutions:

$$\phi(x) = \int \frac{d^4p}{(2\pi)^4}e^{-ipx} \qquad \phi^*(x) = \int \frac{d^4p}{(2\pi)^4}e^{ipx}$$

Commutation Relations (Bosonic):

$$[\phi_x, \pi_y] = i\delta^3(\vec{x}-\vec{y}) \qquad [\phi_x,\phi_y] = [\pi_x,\pi_y] = 0$$
$$\left[a_{\vec{k}}, a^\dagger_{\vec{p}}\right] = \left[b_{\vec{k}}, b^\dagger_{\vec{p}}\right] = i(2\pi)^3\delta^3(\vec{k}-\vec{p})$$
$$\left[a_{\vec{k}}, a_{\vec{p}}\right] = \left[a^\dagger_{\vec{k}}, a^\dagger_{\vec{p}}\right] = \left[b_{\vec{k}}, b_{\vec{p}}\right] = \left[b^\dagger_{\vec{k}}, b^\dagger_{\vec{p}}\right] = 0$$
$$\left[H, a_{\vec{k}}\right] = -\omega_{\vec{k}}a_{\vec{k}} \qquad \left[H, a^\dagger_{\vec{k}}\right] = \omega_{\vec{k}}a^\dagger_{\vec{k}} \qquad \left[H, b_{\vec{k}}\right] = -\omega_{\vec{k}}b_{\vec{k}} \qquad \left[H, b^\dagger_{\vec{k}}\right] = \omega_{\vec{k}}b^\dagger_{\vec{k}}$$

Mode Expansion

$$\phi = \int \frac{d^3p}{(2\pi)^3\sqrt{2\omega_p}}\left(a_{\vec{p}}e^{ipx} + b^\dagger_{\vec{p}}e^{-ipx}\right) \qquad \phi^* = \int \frac{d^3p}{(2\pi)^3\sqrt{2\omega_p}}\left(b_{\vec{p}}e^{ipx} + a^\dagger_{\vec{p}}e^{-ipx}\right)$$
$$\pi = i\int \frac{d^3p}{(2\pi)^3}\sqrt{\frac{\omega_p}{2}}\left(-b_{\vec{p}}e^{ipx} + a^\dagger_{\vec{p}}e^{-ipx}\right) \qquad \pi^* = i\int \frac{d^3p}{(2\pi)^3}\sqrt{\frac{\omega_p}{2}}\left(-a_{\vec{p}}e^{ipx} + b^\dagger_{\vec{p}}e^{-ipx}\right)$$
$$H = \int \frac{d^3p}{(2\pi)^3}\omega_p\left(a^\dagger_{\vec{p}}a_{\vec{p}} + b^\dagger_{\vec{p}}b_{\vec{p}} + \frac{1}{2}(2\pi)^3\delta^3(\vec{p}-\vec{p})\right)$$

Bonus - ladder operator as a function of ϕ and π:

$$a_k = \int\left(\sqrt{\frac{\omega_k}{2}}\phi + \frac{i}{\sqrt{2\omega_k}}\pi^*\right)e^{-kx} \qquad b_k = \int\left(\sqrt{\frac{\omega_k}{2}}\phi^* + \frac{i}{\sqrt{2\omega_k}}\pi\right)e^{-kx}$$

4.3 Weyl Spinor Field

This section treats Left Handed Weyl Spinors. To change to Right Handed spinors, replace $\bar{\sigma} \leftrightarrow \sigma$.
Lagrangian:

$$\mathscr{L} = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi \qquad \psi \in \mathbb{C}^2 \qquad \bar{\sigma} = (\mathbb{1}, -\vec{\sigma})$$

Momentum and Hamiltonian:

$$\mathscr{H} = i\psi^\dagger \vec{\sigma} \cdot \nabla \psi \qquad \pi = i\psi^\dagger$$

Equation of Motion - Weyl equation

$$\bar{\sigma}^\mu \partial_\mu \psi = 0$$

Applying $\sigma_\nu \partial^\nu$ to both sides gives the massless KG equation $\Box \psi = 0$.
Solutions:

$$\psi(x) = \xi_p e^{\pm px} \qquad \xi_p = \sqrt{2\omega_p} U \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \sqrt{\sigma p} U \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \sqrt{\sigma p} = \frac{\sigma p}{\sqrt{2\omega_p}}$$

Where $U \in SU(2)$ and ξ are the helicity eigenfunctions:

$$\sigma_i \hat{p}^i \xi_p = -p \xi_p$$

For RH Weyl Spinor, the eigenvalue is positive.
Commutation Relations (Fermionic):

$$\left\{ \psi_a(\vec{x}), \psi_b^\dagger(\vec{y}) \right\} = \delta^3(\vec{x} - \vec{y}) \delta_{ab} \qquad \{ \psi_a(\vec{x}), \psi_b(\vec{y}) \} = \left\{ \psi_a^\dagger(\vec{x}), \psi_b^\dagger(\vec{y}) \right\} = 0$$

$$\left\{ b_k, b_p^\dagger \right\} = \left\{ c_k, c_p^\dagger \right\} = (2\pi)^3 \delta(\vec{k} - \vec{p}) \qquad \left\{ b_k, b_p \right\} = \left\{ b_k^\dagger, b_p^\dagger \right\} = \left\{ c_k, c_p \right\} = \left\{ c_k^\dagger, c_p^\dagger \right\} = \left\{ b_k, c_p \right\} = \left\{ b_k^\dagger, c_p^\dagger \right\} = 0$$

Mode expansion:

$$\psi = \int \frac{d^3k}{(2\pi)^3 \sqrt{2\omega_k}} \left(\xi(\vec{k}) b_{\vec{k}} + \xi(-\vec{k}) c_{\vec{k}}^\dagger \right) e^{i\vec{k} \cdot \vec{x}}$$

$$H = \int \frac{d^3k}{(2\pi)^3} \omega_k \left(b_{\vec{k}}^\dagger b_k + c_k^\dagger c_{\vec{k}} \right)$$

4.4 Dirac Spinor

The Dirac Spinor:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \qquad \bar{\psi} = \psi^\dagger \gamma^0 = \begin{pmatrix} \psi_L^\dagger & \psi_R^\dagger \end{pmatrix}$$

Lagrangian:

$$\mathscr{L} = \bar{\psi} \left(i\partial\!\!\!/ - m \right) \psi$$

Momentum and Hamiltonian:

$$\mathscr{H} = \bar{\psi} \left(-i\gamma^i \partial_i + m \right) \psi \qquad \pi = i\psi^\dagger$$

Equation of Motion - Dirac equation

$$\left(i\partial\!\!\!/ - m \right) \psi = 0$$

Applying $\left(i\partial\!\!\!/ + m \right)$ to both sides gives the massive KG equation $\left(\Box + m^2 \right) \psi = 0$.
Solutions:

$$\psi(x) = u_s(p) e^{-px} \qquad u_s(p) = \begin{pmatrix} \sqrt{\sigma p} \xi_s \\ \sqrt{\bar{\sigma} p} \xi_s \end{pmatrix} \qquad \sqrt{\sigma p} = \frac{\mathbb{1} \left(p^0 + m \right) - \sigma_i p^i}{\sqrt{p^0 + m}} \qquad \sqrt{\sigma p} = \frac{\mathbb{1} \left(p^0 + m \right) + \sigma_i p^i}{\sqrt{p^0 + m}} \qquad s = \pm \frac{1}{2}$$

For the opposite equation:

$$\left(i\partial\!\!\!/ + m \right) \psi = 0$$

The solutions are:

$$\psi(x) = v_s(p)e^{-px} \qquad v_s(p) = \begin{pmatrix} \sqrt{\sigma p}\xi_{-s} \\ -\sqrt{\bar{\sigma} p}\xi_{-s} \end{pmatrix} \qquad \xi_{-s} = -i\sigma_2 \xi_s^*$$

Identities:

$$\bar{u}_s u_r = 2m\delta_{sr} \qquad u_s^\dagger u_r = 2p^0 \delta_{sr} \qquad \sum_s u_s \bar{u}_s = \not{p} + m$$

$$\bar{v}_s v_r = -2m\delta_{sr} \qquad v_s^\dagger v_r = 2p^0 \delta_{sr} \qquad \sum_s v_s \bar{v}_s = \not{p} - m$$

$$\bar{u}_s v_r = \bar{v}_s u_s = 0$$

Commutation Relations (Fermionic):

$$\left\{\psi_a(\vec{x}), \psi_b^\dagger(\vec{y})\right\} = \delta^3(\vec{x}-\vec{y})\delta_{ab} \qquad \{\psi_a(\vec{x}), \psi_b(\vec{y})\} = \left\{\psi_a^\dagger(\vec{x}), \psi_b^\dagger(\vec{y})\right\} = 0$$

$$\left\{a_{\vec{p},s}, a_{\vec{k},r}^\dagger\right\} = \left\{b_{\vec{p},s}, b_{\vec{k},r}^\dagger\right\} = (2\pi)^3\delta(\vec{k}-\vec{p})\delta_{sr}$$

$$\left\{a_k, a_p\right\} = \left\{a_k^\dagger, a_p^\dagger\right\} = \left\{b_k, b_p\right\} = \left\{b_k^\dagger, b_p^\dagger\right\} = \left\{a_k, b_p\right\} = \left\{a_k^\dagger, b_p^\dagger\right\} = 0$$

Mode expansion:

$$\psi = \int \frac{d^3p}{(2\pi)^3\sqrt{2\omega_p}} \sum_s \left(a_{\vec{p},s}u_s(p)e^{-ipx} + b_{\vec{p},s}^\dagger v_s(p)e^{ipx}\right) \qquad \bar{\psi} = \int \frac{d^3p}{(2\pi)^3\sqrt{2\omega_p}} \sum_s \left(a_{\vec{p},s}^\dagger \bar{u}_s(p)e^{ipx} + b_{\vec{p},s}\bar{v}_s(p)e^{-ipx}\right)$$

$$H = \int \frac{d^3k}{(2\pi)^3}\omega_k \left(a_{\vec{k}}^\dagger a_k + b_k^\dagger b_{\vec{k}}\right)$$

5 Interacting theories

5.1 Quantum Electrodynamics

Lagrangian:

$$\mathscr{L} = \bar{\psi}\left(i\not{D} - m\right)\psi = \bar{\psi}\left(i\not{\partial} - m - e\not{A}\right)\psi \qquad D_\mu = \partial_\mu + ieA_\mu$$

Gauge redundancy:

$$\begin{Bmatrix} A_\mu \to A_\mu - \frac{1}{e}\partial_\mu\alpha(x) \\ \psi \to e^{i\alpha(x)}\psi \end{Bmatrix} \Rightarrow \bar{\psi}\not{D}\psi \to \bar{\psi}\not{D}\psi$$

The Interaction terms:

$$\mathscr{L}_I = -e\bar{\psi}\not{A}\psi \qquad H_I = e\int d^3x\bar{\psi}\not{A}\psi$$

First order (non-reletavistic):

$$H_I\left|ps\right\rangle \supset e\tilde{A}_0\left|ps\right\rangle$$

Second order (First order Magentic Interaction) splits into spin-diagonal term:

$$H_I\left|ps\right\rangle \supset e\int \frac{d^3k}{(2\pi)^3\sqrt{2\omega_k}}(p+k)^j\delta_{rs}a_{\vec{k},r}^\dagger\left|0\right\rangle\tilde{A}_j(\vec{k}-\vec{p})$$

And spin dependent term:

$$H_I\left|ps\right\rangle \supset -\frac{e}{m}\vec{B}\cdot\vec{S}_{rs}\left|ps\right\rangle$$

5.2 Interactions & Feynman Diagrams

Note (out of material): The limit on interactions comed from the coupling being allowed only non-negative dimension.
The Interaction picture:

$$H(\phi(t)) = H_0(\phi(t)) + H_I(\phi(t))$$

The full field is then:

$$\phi(t) = U^\dagger(t,t_0)\phi_I(t)U(t,t_0) \qquad \begin{aligned} \phi_I(t) &= e^{iH_0(\phi(t_0))(t-t_0)}\phi(t_0)e^{-iH_0(\phi(t_0))(t-t_0)} \\ U(t,t_0) &= e^{iH_0H(\phi(t_0))(t-t_0)}e^{-iH(\phi(t))(t-t_0)} \end{aligned}$$

The propeagtor $U(t,t_0)$ follows:

$$i\partial_t U(t,t_0) = H_I(\phi_I(t))U(t,t_0)$$

Solution - Dyson's Formula ($H_I(t) = H_I(\phi_I(t))$):

$$U(t,t_0) = 1 + \int_{t_0}^{t} dt'(-i)H_I(t') + \int_{t_0}^{t} dt'(-i)H_I(t')\int_{t_0}^{t'} dt''(-i)H_I(t'') + \cdots = T\left\{\exp\left[-i\int_{t_0}^{t} d\tilde{t}H_I(\tilde{t})\right]\right\}$$

Generally:

$$U_I(t,t') = T\left\{\exp\left[-i\int_{t'}^{t} d\tilde{t}H_I(\tilde{t})\right]\right\} = e^{iH_0(\phi(t_0))(t-t_0)}e^{-iH(\phi(t))(t-t')}e^{-iH_0(\phi(t_0))(t'-t_0)}$$

Decomposition:

$$U_I(t_3,t_2)U_I(t_2,t_1) = U_I(t_3,t_1)$$

Hermitian Conjugate is backwards propegator:

$$U_I^\dagger(t_2,t_1) = U_I^{-1}(t_2,t_1) = U_I(t_1,t_2)$$

Ground state:

$$|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)}\left[e^{-\varepsilon_0(t_0-(-T))}\langle\Omega|0\rangle\right]^{-1}U_I(t_0,-T)|0\rangle$$

$$\langle\Omega| = \lim_{T\to\infty(1-i\varepsilon)}\left[e^{-\varepsilon_0(T-t_0)}\langle 0|\Omega\rangle\right]^{-1}\langle 0|U_I(T,t_0)$$

$$H|\Omega\rangle = \varepsilon_0|\Omega\rangle$$

Where ε_0 is the vacuum energy:

$$e^{-iHt}|0\rangle = e^{-i\varepsilon_0 t}\langle\Omega|0\rangle + \sum_n e^{-i\varepsilon_n t}\langle n|0\rangle$$

Correlation Function:

$$\langle\Omega|T\{\phi_1\phi_2\cdots\phi_N\}|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)}\frac{\langle 0|T\{\phi_{I,1}\phi_{I,2}\cdots\phi_{I,N}U(T,-T)\}|0\rangle}{\langle 0|U(T,-T)|0\rangle}$$

Wick's theorem: for N fields $\phi_1,\ldots,\phi_N$:

$$\begin{gathered} T\{\phi_1,\ldots,\phi_N\} = N\{\phi_1,\ldots,\phi_N\} + \\ +N\left\{\phi_1\phi_2\phi_3\cdots\phi_N + \phi_1\phi_2\phi_3\cdots\phi_N + \cdots + \phi_1\phi_2\phi_3\cdots\phi_N\right\} + \\ +N\left\{\phi_1\phi_2\phi_3\phi_4\cdots\phi_N + \phi_1\phi_2\phi_3\phi_4\cdots\phi_N + \cdots\right\} \end{gathered}$$

Drawing Feynman Diagrams:

- Start from the interacting term $\mathscr{L}_I$

- Decide on N the amount of external points (particles you start with)
- Expand the correlation function using Dyson's formula up to the desired order in the coupling variable.
- For each term:
  - Go over all possible contraction perturbations, and count the amount of each type.
  - To draw the diagrams, draw points for each external fields at the start and end of the interaction, then connect them by the contractions. If tehre are additional internal fields (interactions), draw them as a point in the middle and add their interactions.
  - Discard all diagrams that have vacuum loops.
  - To write in integral form, for each perturbation topology, write the amount as a prefactor and a Feynman propegator for each contraction. If the propegator has an internal field, it retains the integral over all possible points.

5.3 2-point correlation function

These are the simplest building blocks. The free propeagtor

$$D(x-y)=\int\frac{d^3k}{(2\pi)^3 2\omega_k}e^{-ik(x-y)}$$

The Feynman propeagtor:

$$D_F(x-y)=\langle 0|T\phi(x)\phi(y)|0\rangle=\Theta(x_0-y_0)D(x-y)+\Theta(y_0-x_0)D(y-x)=\lim_{\varepsilon\to 0^+}\int\frac{d^4k}{(2\pi)^4}\frac{i}{k^2-m^2+i\varepsilon}e^{-ik(x-y)}$$

With interaction H_I:

$$T\left\{\phi(x)\phi(y)\right\}=N\left\{\phi(x)\phi(y)\right\}+\wick{\c\phi(x)\c\phi(y)}$$

Where $\wick{\c\phi(x)\c\phi(y)}$ is the contraction:

$$\wick{\c\phi(x)\c\phi(y)}=D_F(x-y)=\begin{cases}[\phi_+(x),\phi_-(y)] & x_0>y_0\\ [\phi_+(y),\phi_-(x)] & y_0>x_0\end{cases}$$

Example - if the interaction term is:

$$\mathscr{L}_I=\frac{\lambda}{4!}\phi^4$$

The Correlation function up to first order:

$$\langle\Omega|T\phi_x\phi_y|\Omega\rangle=\langle 0|T\phi_x\phi_y|0\rangle+\int d^4z\left(\frac{-i\lambda}{4!}\right)\langle 0|T\phi_x\phi_y\phi_z^4|0\rangle=$$
$$=D_F(x-y)-\frac{\lambda}{4!}\left(3D_F(x-y)\int d^4zD_F^2(z-z)+3\cdot 4\int d^4zD_F(x-z)D_F(y-z)D_F(z-z)\right)$$

5.4 Scattering Matrix and Observables

The scattering matrix connects asymptotic free states:

$$|\text{out}\rangle=S|\text{in}\rangle$$

Defined as:

$$S=\lim_{T\to\infty}U_I(T,-T)$$

with:

$$S = \mathbb{1} + iT$$

LSZ Reduction Formula: Scattering amplitudes are obtained from time-ordered correlators:

$$\langle p_1 \cdots p_n | S | k_1 \cdots k_m \rangle \propto \prod_i (\Box_{x_i} + m^2) \langle \Omega | T\phi(x_1) \cdots \phi(x_N) | \Omega \rangle$$

External legs are amputated propagators.
The Matrix elements are

$$\langle f | S | i \rangle = i(2\pi)^4 \delta^{(4)}(p_f - p_i) M$$

M is computed from Feynman diagrams.
Cross section of $2 \to n$ scattering:

$$d\sigma = \frac{1}{4E_a E_b |\vec{v}_1 - \vec{v}_2|} \prod_f \frac{d^3 p_f}{(2\pi)^3} \frac{1}{2E_f} \left| M(\to \{p_f\}) \right|^2 (2\pi)^4 \delta^4 \left(p_a + p_b - \sum p_f \right)$$

Decay rate of a particle:

$$d\Gamma = \frac{1}{2m_i} \frac{d^3 p_f}{(2\pi)^3} \frac{1}{2E_f} \left| M(\to \{p_f\}) \right|^2 (2\pi)^4 \delta^4 \left(p_a + p_b - \sum p_f \right)$$

6 Continous Symmetries & Noether's theorem

6.1 Finding currents

To use Neother's theorem:

- Parameterize the symmetry transformation's action on the fields.
- Derive $\delta\phi$ - the infinitesimal change in the field.
- Promote the transformation parameter to a function of spacetime.
- Insert into δS or $\delta\mathscr{L}$.
- The result should give:

$$\delta S = \int d^4 x \delta\mathscr{L} \qquad \delta\mathscr{L} = \varepsilon \partial_\mu J^\mu$$

- Might need to use integration by parts on the action integral to get there.
- J^μ are the currents.

Another way to do it is to derive ϕ', which is the field after the transformation with parameter ε, and

$$\delta\mathscr{L} = L(\phi') - L(\phi) = \varepsilon \partial_\mu F^\mu$$

For some F^μ. Then, the current is:

$$J_\mu = \frac{\partial\mathscr{L}}{\partial\left(\partial^\mu \phi\right)} \frac{\partial\phi'}{\partial\varepsilon} - F_\mu$$

7 Discrete Symmetries

7.1 Parity P

Parity acts as a spatial inversion:

$$P: \quad x^\mu = (t,\vec{x}) \to (t,-\vec{x})$$

For fields:

$$\phi(t,\vec{x}) \to \eta_P \phi(t,-\vec{x})$$

$$\psi(t,\vec{x}) \to \eta_P \gamma^0 \psi(t,-\vec{x})$$

7.2 Charge Conjugation C

Charge conjugation exchanges particles and antiparticles.

$$C: \quad \psi \to \psi^C = C\bar{\psi}^T$$

with

$$C\gamma^\mu C^{-1} = -(\gamma^\mu)^T$$

For gauge fields:

$$A_\mu \to -A_\mu$$

7.3 Time Reversal T

Time reversal reverses time evolution:

$$T: \quad (t,\vec{x}) \to (-t,\vec{x})$$

It is anti-unitary:

$$TiT^{-1} = -i$$

7.4 CPT Theorem

Any local, Lorentz-invariant, unitary quantum field theory satisfies:

$$\mathscr{L}(x) \xrightarrow{CPT} \mathscr{L}(-x)$$

Consequences:

- Particles and antiparticles have equal masses and lifetimes
- Spin–statistics consistency
- Violation of C, P, or T individually does not imply CPT violation

8 Groups and Representations

8.1 The Lorentz group

The Energy Momentum Tensor:

$$T_{\mu\nu} = \eta_{\mu\nu}\mathscr{L} - \partial_\mu\phi\partial_\nu\phi$$

From the spacetime translation symmetry, the conserved currents are:

$$J_{(\alpha\beta)\mu} = x_\alpha T_{\beta\mu} - x_\beta T_{\alpha\mu}$$

This gives six currents corrsponding to three rotations and three boosts. The charges (and currents) follow the commutation relations:

$$\left[Q_{\mu\nu}, Q_{\alpha\beta}\right] = i\left(\eta_{\alpha\nu}Q_{\mu\beta} + \eta_{\alpha\mu}Q_{\beta\nu} + \eta_{\beta\nu}Q_{\alpha\mu} + \eta_{\beta\mu}Q_{\nu\alpha}\right) \tag{1}$$

And they are named:

$$L_1 = Q_{23} \qquad L_2 = Q_{31} \qquad L_3 = Q_{12} \qquad M_i = Q_{0i}$$

The four-vector representation:

$$[J^{\mu\nu}]^{\lambda}_{\sigma} = i\left(\delta^{\mu k}\delta^{\nu}_{\sigma} - \delta^{\mu}_{\sigma}\delta^{\nu k}\right)\eta^{\lambda k}$$

The two-vector representations:

$$J_i^{\pm} = \frac{1}{2}\left(L_i \pm iM_i\right)$$

Weyl spinnor transformation a_α is responsible for rotation and b_α is responsible for boosts:

$$\psi_L(x) \to \psi_L'(x) = e^{-i(a_\alpha - ib_\alpha)\sigma_\alpha/2}\psi_L(\Lambda^{-1}x)$$
$$\psi_L(x) \to \psi_L'(x) = e^{-i(a_\alpha + ib_\alpha)\sigma_\alpha/2}\psi_L(\Lambda^{-1}x)$$

8.2 Pauli matrices

$$\sigma_0 = \mathbb{1} \qquad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Identities:

$$\sigma_i\sigma_j = \delta_{ij}\mathbb{1} + i\varepsilon_{ijk}\sigma_k \qquad \sigma_i^\dagger = \sigma_i \qquad [\sigma_i, \sigma_j] = 2i\sigma_l \qquad \left\{\sigma_i, \sigma_j\right\} = 2\delta_{ij}\mathbb{1}$$
$$e^{i\vec{\theta}\cdot\vec{\sigma}} = \cos\theta\mathbb{1} + i\sin\theta\hat{\theta}\cdot\vec{\sigma} \qquad \vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$$
$$\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = 2\eta^{\mu\nu} \qquad \sigma^\mu = (\sigma_0, \vec{\sigma}) \qquad \bar{\sigma}^\mu = (\sigma_0, -\vec{\sigma})$$
$$\sigma_i^T\sigma_2 = -\sigma_2\sigma_i \qquad \sigma_2\bar{\sigma}^\mu\sigma_2 = (\sigma^\mu)^T$$

8.3 Gamma Matrices

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \qquad \not{A} = \gamma^\mu A_\mu$$

Identities:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{1}_4 \qquad \not{a}\not{b} + \not{b}\not{a} = 2ab\mathbb{1} \qquad \not{a}\not{a} = a^2\mathbb{1}$$